SILVER ZONE FOUNDATION
NEW DELHI, INDIA

Do not start attempting the test paper until you are asked to do so.

ENROLLMENT NUMBER:

STUDENT NAME:

Class 6

Maximum Marks : 100
Duration : 40 Minutes
Total Question : 50
Test Paper Type : A D

Level - 2

iOS 9th INTERNATIONAL OLYMPIAD OF SCIENCE 2017-2018

Instructions for the Candidates

1. Write your 12 digit iOS Enrollment Number and your name on top of the question Paper in the given space.
2. Filling up improper Enrollment Number may lead to unavailability of Result.
3. This Question Paper cum Answer Sheet must be submitted to your teacher after the completion of the test.
4. Tick ☑ your answer (A, B, C, D or E) on the given Box ☐ in this Question Paper cum Answer Sheet with HB Pencil or Black/Blue Ball point Pen.
5. The Question Paper Cum Answer sheet Booklet consists of 50 questions, divided into three sections.
6. Section-A: Science (30 questions); Section-B: Mental Ability and Reasoning (10 questions); Section –C: Scholar's Zone (10 questions).
7. Each question of Section-A carries 1.5 marks; Section-B, 2.5 marks each; and that of Scholar's Zone, 3 marks each.
8. Use of calculator or any other devices in the examination is strictly prohibited.
9. This question paper contains 8 pages.
10. Rough work should be done on the blank sheet/space provided in the question paper booklet.
11. Do not start attempting the test paper until you are asked to do so.
12. All the questions are compulsory and there is no negative marking.

SIGN. OF THE TEACHER

SIGN. OF THE STUDENT

1. Which mineral among the following is essential for the proper functioning of thyroid gland?

   A. ☐ Phosphorus
   B. ☐ Iron
   C. ☐ Iodine
   D. ☐ Potassium
   E. ☐ None of these

2. Which one of the following flowers are eaten as food?

   A. ☐ Sunflower B. ☐ Dandelion
   C. ☐ Gladiolus D. ☐ All the above
   E. ☐ None of these

3. Which one of the following statements is correct?

   A. ☐ Hand spindle (takli) is a traditional way of weaving.
   B. ☐ Two sets of yarns are arranged together in different ways to get different patterns is known as weaving.
   C. ☐ Looms are used in the process of spinning.
   D. ☐ Ginning involves removing of the wool from sheep by using special clippers.
   E. ☐ None of these

4. In which one of the following groups the metal "gold" should be categorised?

   A. ☐ Element B. ☐ Compound
   C. ☐ Mixture D. ☐ All the above
   E. ☐ None of these

5. Which one of the following is not a characteristic of the image formed by a pinhole camera?

   A. ☐ Image is real.
   B. ☐ Image is of same colour as the object.
   C. ☐ Image is always smaller than the object.
   D. ☐ Image is inverted.
   E. ☐ None of these

6. Identify the object from the following which cannot be used for the process of filtration.

   A. ☐ Wire-mesh
   B. ☐ Piece of muslin cloth
   C. ☐ Strainer
   D. ☐ A layer of sand
   E. ☐ None of these

7. Which one of the following parts of our body does not contain cartilage?

   A. ☐ Pinnae of ear
   B. ☐ End of nose
   C. ☐ Ear lobe
   D. ☐ Between the vertebrae of backbone
   E. ☐ None of these

8. Height of a boy is 2 metres and 9 centimetres. Convert his height into decimetre.

   A. ☐ 1.65 dm B. ☐ 20.9 dm
   C. ☐ 165 dm D. ☐ 17.9 dm
   E. ☐ None of these

9. Which one of the following statements is incorrect about magnets?

   A. ☐ Magnets are used to separate discarded iron and steel objects from other waste materials.
   B. ☐ Magnetism of a magnet can be increased by heating it over a burner.
   C. ☐ A magnet should never be hit with a hammer to retain its magnetism.
   D. ☐ A magnet should not be dropped on the floor to retain its magnetism.
   E. ☐ None of these

10. Which one of the following parts of plant contains male sex cells?

    A. ☐ Ovule B. ☐ Filament
    C. ☐ Anther D. ☐ Pistil
    E. ☐ None of these

11. Which one of the following is/are not always necessary to observe a shadow?

    A. ☐ Sun
    B. ☐ Screen
    C. ☐ Source of light
    D. ☐ Opaque object
    E. ☐ None of these

12. Presence of increased _____ in the air means increased humidity.

    A. ☐ dust particles
    B. ☐ oxygen
    C. ☐ water vapor
    D. ☐ smoke
    E. ☐ None of these

13. Which one of the following statements is correct?

A. ☐ When the speed of an object is doubled, the kinetic energy possessed by the object is increased by four times.

B. ☐ When the speed of an object is doubled, the kinetic energy possessed by the object is increased by two times.

C. ☐ Kinetic energy of an object does not depend upon the speed of the object.

D. ☐ All the above

E. ☐ None of these

14. Which one of the following is a gastropod?

A. ☐ 

B. ☐ 

C. ☐ 

D. ☐ 

E. ☐ None of these

15. In which one of the following parts of the digestive tract, digested food is absorbed by the blood vessels?

A. ☐ Large intestine

B. ☐ Buccal cavity

C. ☐ Stomach

D. ☐ Small intestine

E. ☐ None of these

16. When electricity from the cell passes through the filament of the bulb, the bulb lightens up due to:

A. ☐ Heating effect of the electric current

B. ☐ Chemical effect of the electric current

C. ☐ Magnetic effect of the electric current

D. ☐ All the above

E. ☐ None of these

17. _____ fibre is also known as artificial silk.

A. ☐ Cotton B. ☐ Rayon

C. ☐ Polyester D. ☐ Coir

E. ☐ None of these

18. Hand picking method is effective for which mixture?

A. ☐ Mixture of gas in gas

B. ☐ Mixture of solid in solid

C. ☐ Mixture of liquid in liquid

D. ☐ All the above

E. ☐ None of these

19. Liquid waste collected from houses is known as:

A. ☐ Effluent B. ☐ Smoke

C. ☐ Smog D. ☐ Sewage

E. ☐ None of these

20. What are the factors on which the shape of the shadow does not depend?

A. ☐ Shape of the object

B. ☐ Position of the source of light

C. ☐ Colour of the object

D. ☐ All the above

E. ☐ None of these

21. Choose the incorrect statement from the following.

A. ☐ Saw-dust and dry leaves are lighter than water, thus float on water.

B. ☐ Nitrogen and oxygen get dissolved in water.

C. ☐ Glass and stone are heavier than water, thus float on water.

D. ☐ Wood and rubber have dull appearance.

E. ☐ None of these

22. Motion of the second's hand of a watch is an example of:

A. ☐ Rectilinear motion

B. ☐ Periodic motion

C. ☐ Oscillatory motion

D. ☐ All the above

E. ☐ None of these

23. Iron blade of a spade is made slightly smaller than the wooden handle because:

A. ☐ Iron blade contracts on heating.

B. ☐ Iron blade expands on heating.

C. ☐ Iron blade remains the same on heating.

D. ☐ All the above

E. ☐ None of these

24. _____ is also known as tailbone.

A. ☐ Cervical B. ☐ Coccyx
C. ☐ Sacrum D. ☐ Thoracic
E. ☐ None of these

25. Which plant among the following is both producer and carnivorous?

A. ☐ Neem B. ☐ Croton
C. ☐ Sundew D. ☐ Mango
E. ☐ None of these

26. A car travels a distance of 263 Km in 5 hours. What would be the speed of the car?

A. ☐ 26.5 Km/hr B. ☐ 52.6 Km/hr
C. ☐ 50.6 Km/hr D. ☐ 49.6 Km/hr
E. ☐ None of these

27. Density of an object refers to:

A. ☐ product of mass and volume
B. ☐ addition of mass and volume
C. ☐ mass per unit temperature
D. ☐ mass per unit volume
E. ☐ None of these

28. The speed of sound depends on the physical features of the medium.Which one of the following mediums allows the sound to travel with maximum speed?

A. ☐ Milk B. ☐ Water
C. ☐ Air D. ☐ Iron
E. ☐ None of these

29. An object of mass 10 Kg is moving with the velocity of 5 m/s. Calculate the kinetic energy possessed by the object.

A. ☐ 100 J B. ☐ 12.5 J
C. ☐ 125 J D. ☐ 120 J
E. ☐ None of these

30. If you dump kitchen waste in a pit, it may, after sometime:

A. ☐ convert into vermicompost.
B. ☐ convert into compost.
C. ☐ remain as such.
D. ☐ remain forever in its dried form.
E. ☐ None of these

Section–B (Mental Ability and Reasoning)

31. Complete the series given below.

7, 14, 49, 98, 343, 686, ?

A. ☐ 4802
B. ☐ 2401
C. ☐ 3206
D. ☐ 4082
E. ☐ None of these

32. Pointing to an old man, Ritwik said, " His son is my son's uncle", how is the old man related to Ritwik?

A. ☐ Father
B. ☐ Uncle
C. ☐ Grandfather
D. ☐ Nephew
E. ☐ None of these

33. At my school, I am facing West, then I turn right and go 10 m straight, then turn left and go 20 m straight, then I turn right and go 10 m straight and from there again I turn right and go 20 m straight. In which direction am I from my school?

A. ☐ East B. ☐ South
C. ☐ North D. ☐ West
E. ☐ None of these

34. If EXPLANATION is coded as GVRJCLCRKMP, then APTITUDE will be coded as:

A. ☐ SVCNGFVS
B. ☐ CNVGVSFC
C. ☐ CNGVFCSV
D. ☐ NGCVSUSV
E. ☐ None of these

Directions for Q 35. and 36: Study the following information and answer the questions given below.

Richard, Oliver and Jenilia are studious.

Alen, Richard and Samuel are naughty.

Samuel, Jenilia and Alen are extrovert.

Richard and Oliver are introvert.

35. Which one of the following persons is neither naughty nor introvert?

A. ☐ Richard B. ☐ Oliver

C. ☐ Jenilia D. ☐ Samuel

E. ☐ None of these

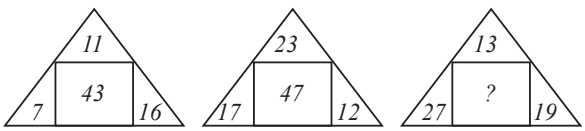
36. Which one of the following persons is neither extrovert nor naughty but is introvert?

A. ☐ Richard B. ☐ Oliver

C. ☐ Samuel D. ☐ Jenilia

E. ☐ None of these

37. Find the missing number.



A. ☐ 59 B. ☐ 51

C. ☐ 49 D. ☐ 60

E. ☐ None of these

38. In a certain code language, 'beautiful girl' is written as 'su ko' and 'beautiful name' is written as 'po su', then code for 'girl' is:

A. ☐ ko

B. ☐ po

C. ☐ su

D. ☐ lo

E. ☐ None of these

39. If $9 \times 5 = 42$, $12 \times 16 = 84$, $51 \times 49 = 300$, then $91 \times 73 = ?$

A. ☐ 495

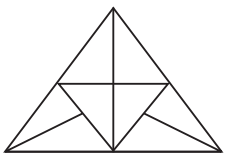
B. ☐ 6643

C. ☐ 492

D. ☐ 54

E. ☐ None of these

40. Count the number of triangles in the figure given below.



A. ☐ 15 B. ☐ 17

C. ☐ 13 D. ☐ 11

E. ☐ None of these

Space for Rough Work

41. Classify the following materials into water soluble and water insoluble.

Alum, Kerosene, Petrol, Salt, Wax, Copper sulphate, Potassium permanganate, Chalk powder, Saw-dust, Washing soda

A. ☐ Water soluble: Copper sulphate, Alum, Salt, Chalk powder, Petrol
Water insoluble: Saw-dust, Kerosene, Potassium permanganate, Wax, Washing soda

B. ☐ Water soluble: Saw-dust, Salt, Chalk powder, Petrol, Washing soda
Water insoluble: Kerosene, Potassium permanganate, Wax, Alum, Copper sulphate

C. ☐ Water soluble: Saw-dust, Salt, Chalk powder, Washing soda, Copper sulphate
Water insoluble: Kerosene, Petrol, Potassium permanganate, Wax, Alum

D. ☐ Water soluble: Alum, Potassium permanganate, Salt, Washing soda, Copper sulphate
Water insoluble: Kerosene, Petrol, Chalk powder, Wax, Saw-dust

E. ☐ None of these

42. Match the following.

	Column A		Column B
I.	*Blue-black colour*	*a.*	*Presence of protein*
II.	*Translucent patch*	*b.*	*Protein deficiency*
III.	*Violet colour*	*c.*	*Presence of starch*
IV.	*Kwashiorkar*	*d.*	*Presence of fat*

A. ☐ I → c, II → d, III → a, IV → b
B. ☐ I → c, II → d, III → b, IV → a
C. ☐ I → d, II → c, III → b, IV → a
D. ☐ I → c, II → b, III → d, IV → a
E. ☐ None of these

43. Which one of the following statements is correct?

I. Threshing is the process which is based on the fact that husk is very light, whereas wheat grains are comparatively heavy.

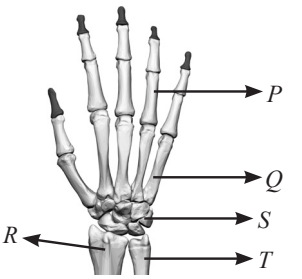
II. Process of decantation can be used for separation only when the solid does not dissolve in the liquid.

III. Sieving is used to separate those solid mixtures which have components of different sizes.

IV. Winnowing is based on the fact that the stalks of the crop plants and the chaff are soft materials, whereas the grains themselves are very hard.

A. ☐ I and II B. ☐ II and III
C. ☐ III and IV D. ☐ I and IV
E. ☐ None of these

44. Choose the correct option to label the diagram given below.



A. ☐ P–Radius, Q–Metacarpals, R–Carpals, S–Phalanges, T–Ulna

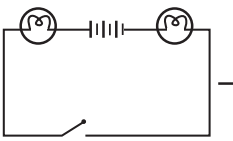
B. ☐ P–Phalanges, Q–Metacarpals, R–Ulna, S–Carpals, T–Radius

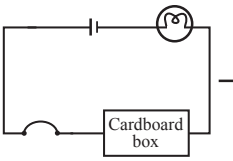
C. ☐ P–Phalanges, Q–Metacarpals, R–Radius, S–Carpals, T–Ulna

D. ☐ P–Metatarcels, Q–Metacarpals, R–Radius, S–Carpals, T– Ulna

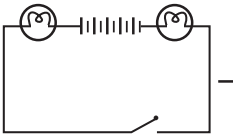
E. ☐ None of these

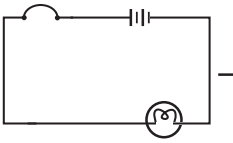
45. Which one of the following is correctly matched?

A. ☐  This circuit consists of two bulbs, a battery of two cells and a closed key

B. ☐ 

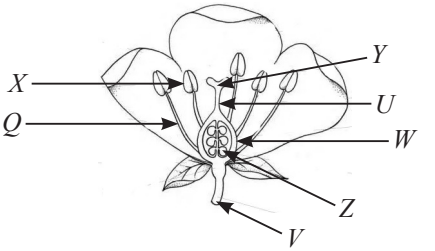
This circuit consists of a battery, one bulb, open key and a conductor

C. ☐  This circuit consists of a battery of five cells, two bulbs and an open key

D. ☐  This circuit consists of a battery of two cells, an open key and an insulator

E. ☐ None of these

46. Choose the correct combination of statements for the structure of flower given below.



I. When pollen grains are transferred from Y to X, it is known as fertilisation.

II. After fertilisation W changes into fruit and Z changes into seeds.

III. When pollen grains are transferred from X to Y, the process is known as pollination.

IV. V is the stalk known as thalamus.

V. Y, U and W are together known as carpel.

A. ☐ I, III and V B. ☐ I, II and III

C. ☐ II, III and IV D. ☐ II, III and V

E. ☐ None of these

47. Himakshi observes a tree through a pinhole camera. Which of the following images depicts the image seen by her correctly?



A. ☐  B. ☐ 

C. ☐  D. ☐ 

E. ☐ None of these

48. Which one of the following animals belongs to the adaptations given below?

I. They have strong flat hind teeth for chewing plants.

II. They have strong legs and can run very fast.

III. They have long ears to hear movements of predators.

IV. They move in herds which also help to protect them from predators.

A. ☐ Octopus and Polar bear

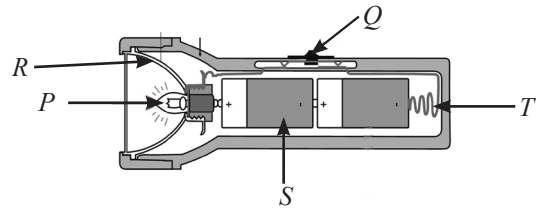
B. ☐ Bison and Zebra

C. ☐ Zebra and Frog

D. ☐ Cow and Penguin

E. ☐ None of these

49. Label the electric torch with the suitable words.



A. ☐ P–Bulb, Q–Slide switch, R–Metal spring, S–Electric cell, T–Reflector

B. ☐ P–Slide switch, Q–Bulb, R–Reflector, S–Electric cell, T–Metal spring

C. ☐ P–Metal spring, Q–Slide switch, R–Bulb, S–Electric cell, T–Reflector

D. ☐ P–Bulb, Q–Slide switch, R–Reflector, S–Electric cell, T–Metal spring

E. ☐ None of these

50. Arek took an empty plastic bottle, turned it upside down and dipped its open mouth into a bucket filled with water. He then tilted the bottle slightly and made the following observations.

i. Bubble of air came out from the bottle.

ii. Some water entered the bottle.

iii. Only nitrogen gas came out in the form of bubbles and oxygen got dissolved in water.

iv. No bubbles were formed, only water entered the bottle.

Which observation is/are correct?

A. ☐ i and ii B. ☐ iv only

C. ☐ iii and iv D. ☐ i only

E. ☐ None of these

Space for Rough Work

ANSWERS

INTERNATIONAL OLYMPIAD OF SCIENCE 2017 (LEVEL 2)

CLASS 6

1.	(C)	2.	(D)	3.	(B)	4.	(A)	5.	(C)	6.	(E)	7.	(C)	8.	(B)	9.	(B)	10.	(C)
11.	(A)	12.	(C)	13.	(A)	14.	(B)	15.	(D)	16.	(A)	17.	(B)	18.	(B)	19.	(D)	20.	(C)
21.	(C)	22.	(B)	23.	(B)	24.	(B)	25.	(C)	26.	(B)	27.	(D)	28.	(D)	29.	(C)	30.	(B)
31.	(B)	32.	(A)	33.	(C)	34.	(B)	35.	(C)	36.	(B)	37.	(B)	38.	(A)	39.	(C)	40.	(B)
41.	(D)	42.	(A)	43.	(B)	44.	(C)	45.	(C)	46.	(D)	47.	(D)	48.	(B)	49.	(D)	50.	(A)